

Project Initialization and Planning Phase

Date	04 July 2026
Team ID	
Project Title	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to develop an intelligent **Human Development Index (HDI) Prediction System** using Machine Learning to accurately predict a country's Human Development Index. The system addresses the limitations of traditional HDI analysis by providing real-time predictions based on health, education, and income indicators. Key features include a Machine Learning-based prediction model, an interactive web interface, country-wise HDI analysis, and fast prediction results.

Project Overview	
Objective	To develop a Machine Learning-based Human Development Index (HDI) Prediction System that accurately predicts HDI using Life Expectancy, Education, and Gross National Income (GNI) indicators, enabling better analysis and decision-making.

Scope	The project focuses on collecting HDI-related data, training a Linear Regression model, and deploying it through a Flask web application for real-time HDI prediction, country comparison, and interactive simulation.
Problem Statement	
Description	Existing HDI analysis methods depend on reports and manual calculations, making it difficult to quickly estimate HDI and understand the impact of changes in health, education, and income indicators.

Impact	An accurate HDI Prediction System helps policymakers, researchers, students, and organizations analyze development levels, compare countries, support policy planning, and make informed decisions..
Proposed Solution	
Approach	Develop a supervised Machine Learning model using historical HDI data and integrate it into a Flask-based web application to generate real-time HDI predictions based on user inputs.
Key Features	- Machine Learning-based HDI prediction model. • Real-time prediction through a web interface. • User-friendly input form for HDI indicators. • Interactive simulation using sliders. • Fast and accurate prediction results. • Country-wise HDI comparison.

	• Explainable Linear Regression model. • Easy deployment using the Flask framework..
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Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computing Resources	Intel Core i5/i7 Processor
	Memory	8 GB RAM or above
	Storage	256 GB SSD
Software	Framework	Flask
	Libraries	scikit-learn, pandas, numpy, matplotlib, seaborn, joblib
	Development Environment	Jupyter Notebook, Visual Studio Code
Data	Data Source	UNDP Human Development Index Dataset
	Size	Approximately 36 Countries (CSV Dataset) <i>(or use your actual dataset size)</i>
	Format	CSV